



MICROECONOMICS

SUPPLY

Chapter 9 | Lecture 2

6 Factors

Individual

$$S_x = f(P, P_r, T, F_p, G_T, G_o)$$

Supply

Market

(6+3)

9 Factors

$$S_m = f(\underbrace{P, P_r, T, F_p, G_T, G_o}_6, \underbrace{F_N, F_U, T_c}_3)$$

SUPPLY FUNCTION

Supply function shows the functional relationship between quantity supplied for a particular commodity and the factors influencing it.

Individual Supply Function

Individual supply function refers to the functional relationship between supply and factors affecting the supply of a commodity.

$$S_x = f(P_x, P_o, P_f, S_t, T, G)$$

Where,

S_x = Market supply of given commodity x;

P_o = Price of other goods;

S_t = State of technology;

G = Goals of the firm.

F = Future expectation regarding P_x

M = Means of transportation and communication.

Market Supply Function

Market supply function refers to the functional relationship between market supply and factors affecting the market supply of a commodity.

$$S_x = f(P_x, P_o, P_f, S_t, T, G, F, M, N)$$

P_x = Price of given commodity x;

P_f = Prices of factors of production;

T = Taxation policy;

N = Number of firms;

SUPPLY SCHEDULE

Supply schedule is a tabular statement showing various quantities of a commodity being supplied at various levels of price, during a given period of time.

Supply → Table



→ 1 Firm

Individual Supply Schedule

Individual supply schedule refers to a tabular statement showing various quantities of a commodity that a producer is willing to sell at various levels of price, during a given period of time.

Price (₹)	Quantity ^① supplied of good x (units)
①	0
②	5
③	10
④	15
⑤	20
⑥	25

P ↑ S ↑ → Profit ↑

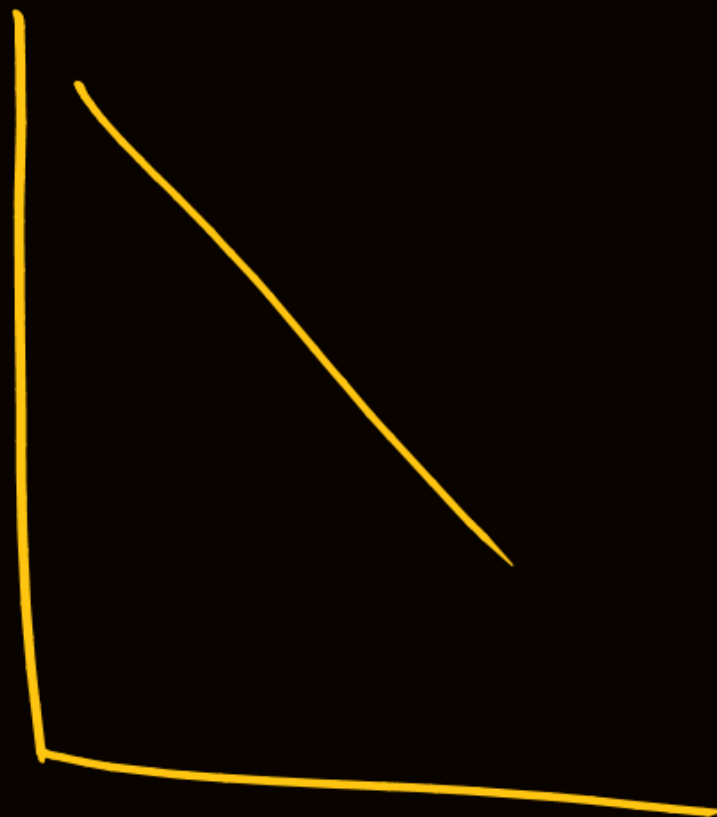
Market Supply Schedule

Market supply schedule refers to a tabular statement showing various quantities of a commodity that all the producers are willing to sell at various levels of price, during a given period of time.

$$S_M = S_A + S_B + \dots\dots\dots$$

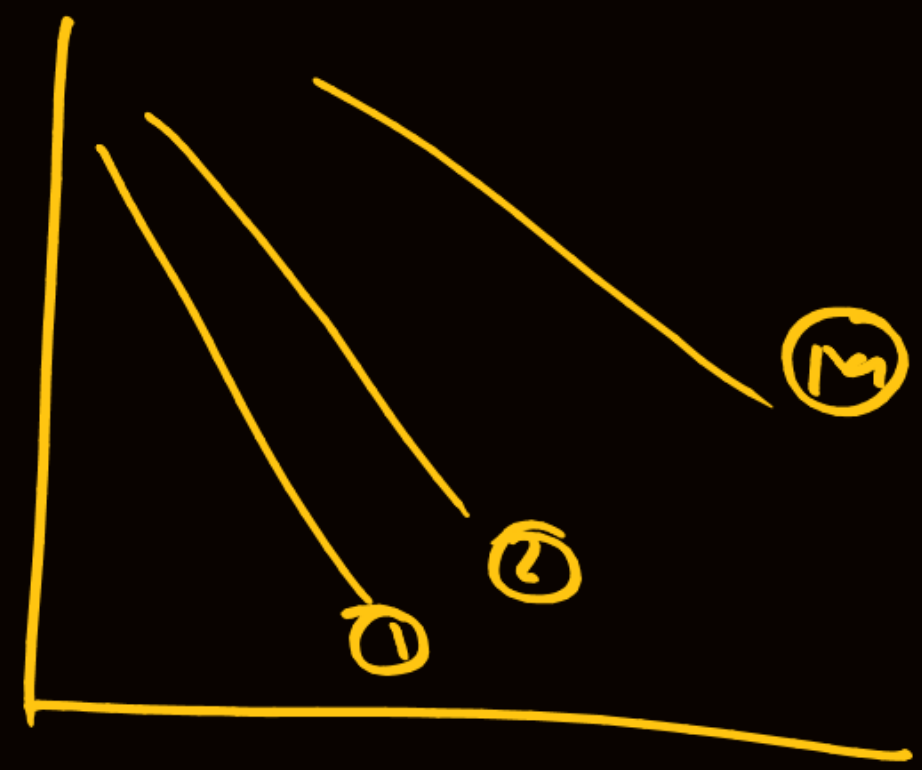
Price (₹) P_x $P \uparrow$	Individual Supply (Units)		Market Supply (units) $(S_A + S_B)$ Total Supply $\uparrow$
	① S_A $S \uparrow$	② S_B $S \uparrow$	
①	0 $\times$	0 $\times$	0
②	5 $\times$	10	5 + 10 = 15
③ $\uparrow$	10 $\uparrow$	20 $\uparrow$	10 + 20 = 30 $\uparrow$
④ $\uparrow$	15 $\uparrow$	25 $\uparrow$	15 + 25 = 40 $\uparrow$
⑤ $\uparrow$	20 $\uparrow$	35 $\uparrow$	20 + 35 = 55 $\uparrow$
⑥ $\uparrow$	25	40 $\uparrow$	25 + 40 = 65 $\uparrow$

Individual Demand



$p \uparrow \Rightarrow D \downarrow$

Market Demand

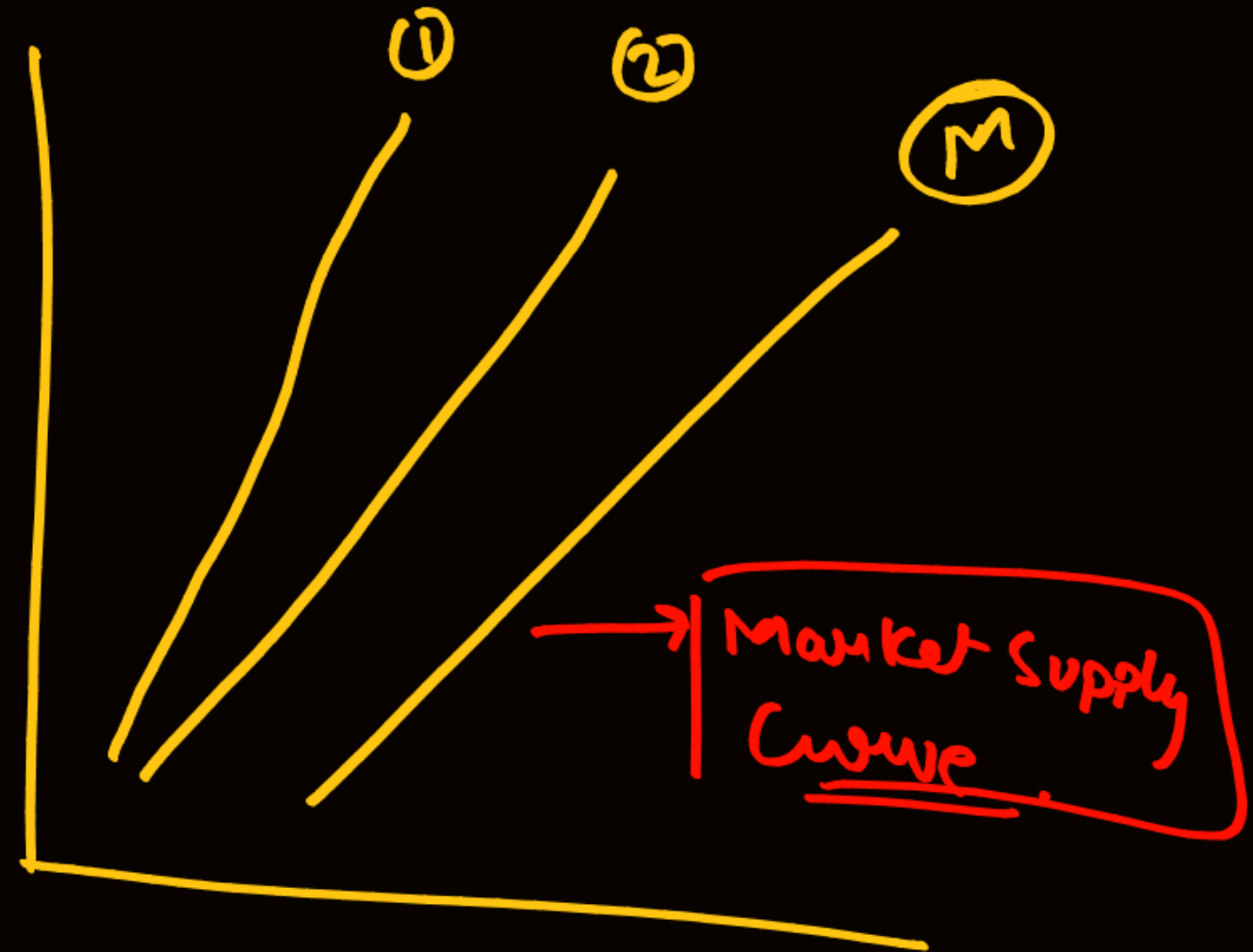


Individual Supply Curve



PTST

Market Supply Curve

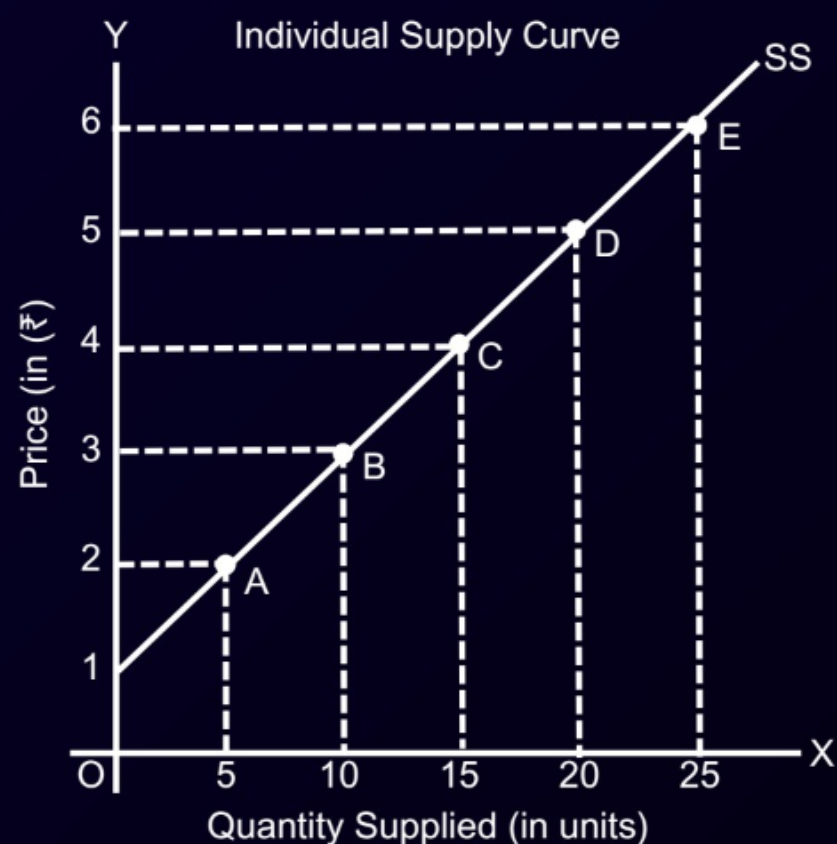


SUPPLY CURVE

Supply curve refers to a graphical representation of supply schedule.

Individual Supply Curve

Individual supply curve refers to a graphical representation of individual supply schedule.



Market Supply Curve

Market supply curve refers to a graphical representation of market supply schedule.

Market supply curve (S_M) is obtained by horizontal summation of the individual supply curves (S_A and S_B).



SUPPLY CURVE

Market Supply Curve in Flatter

Slope of Supply Curve

i.e. Slope of Supply Curve =
$$\frac{\text{Change in Price } (\Delta P)}{\text{Change in Quantity } (\Delta Q)}$$

$$\frac{\Delta P}{\Delta Q}$$

$$\frac{P \uparrow Q \uparrow}{P \downarrow Q \downarrow}$$

Law of Demand

$P \uparrow$ $D \downarrow$

$P \downarrow$ $D \uparrow$



→ Inverse Relation

→ Only Price Change

→ Other Factors Remain Constant

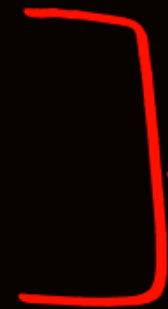
Law of Supply

$P \uparrow$

$S \uparrow$

$P \downarrow$

$S \downarrow$



→ Direct Relation

- only Price change
- keeping other factors constant.

LAW OF SUPPLY

Law of supply states the direct relationship between price and quantity supplied, keeping other factors constant (ceteris paribus).



only price changes

Assumptions of Law of Supply

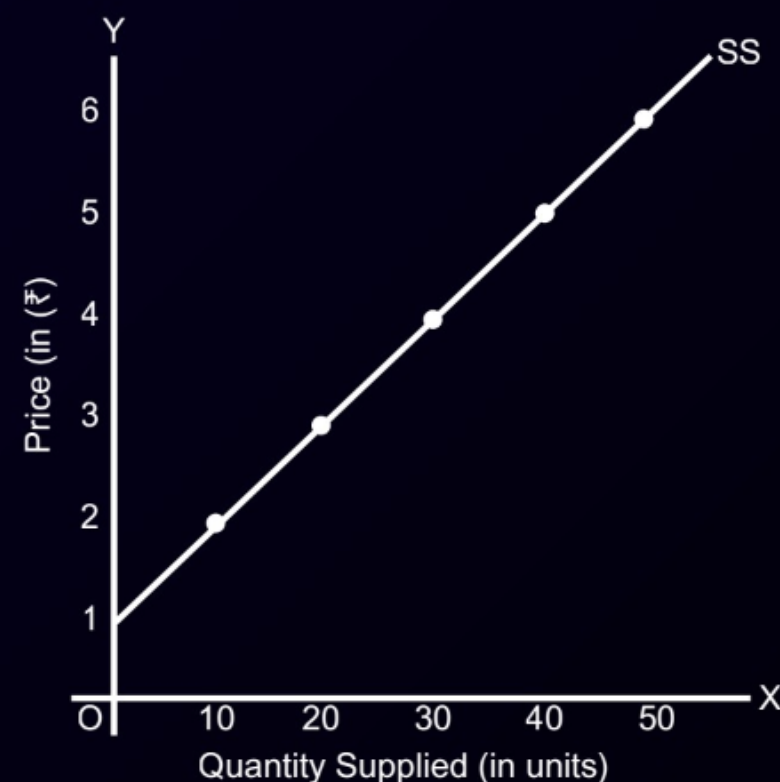
While stating law of supply, the phrase 'keeping other factors constant or ceteris paribus' are used. This phrase is used to cover the following assumptions on which the law is based:

- ✓ Price of other goods is constant;
- ✓ There is no change in the state of technology;
- ✓ Prices of factors of production remain the same;
- ✓ There is no change in the taxation policy;
- ✓ Goals of the producer remain the same.

Price (in ₹)	Quantity (in units)
1	0
2	10
3	20
4	30
5	40
6	50

P↑

S↑



IMPORTANT POINTS ABOUT LAW OF SUPPLY

Important Points about Law of Supply

$P \uparrow S \uparrow \rightarrow$ Direct or Positive

1. It states the **positive relationship** between **price** and quantity supplied, assuming no changes in other factors.
2. It is a **qualitative statement**, as it indicates the **direction** of change in the quantity supplied, but it **does not** indicate the **magnitude** of change.
3. It **does not** establish any **proportional relationship** between change in price and the resultant change in quantity supplied.
4. Law is **one sided** as it explains only the effect of change in price on the supply, and not the effect of change in supply on the price.

elasticity of
Supply

$$E_s = \frac{\Delta Q}{\Delta P} \times \frac{P}{Q}$$

$P \uparrow 50\% \quad S \uparrow 50\% \quad \times$

Reasons for Law of Supply

Profit Motive

The basic aim of producers, while supplying a commodity, is to secure maximum profits. When price of a commodity increases, without any change in costs, it raises their production. On the other hand, with fall in prices, supply also decreases as profit margin decreases at low prices. profits. So, producers increase the supply of the commodity by increasing the

Change in Number of Firms

A rise in price induces the prospective producers to enter into the market to produce the given commodity so as to earn higher profits. Increase in number of firms raises the market supply. However, as the price starts falling, some firms which do not expect to earn any profits at a low price, either stop the production or reduce it. It reduces the supply of the given commodity as the number of firms in the market decreases.

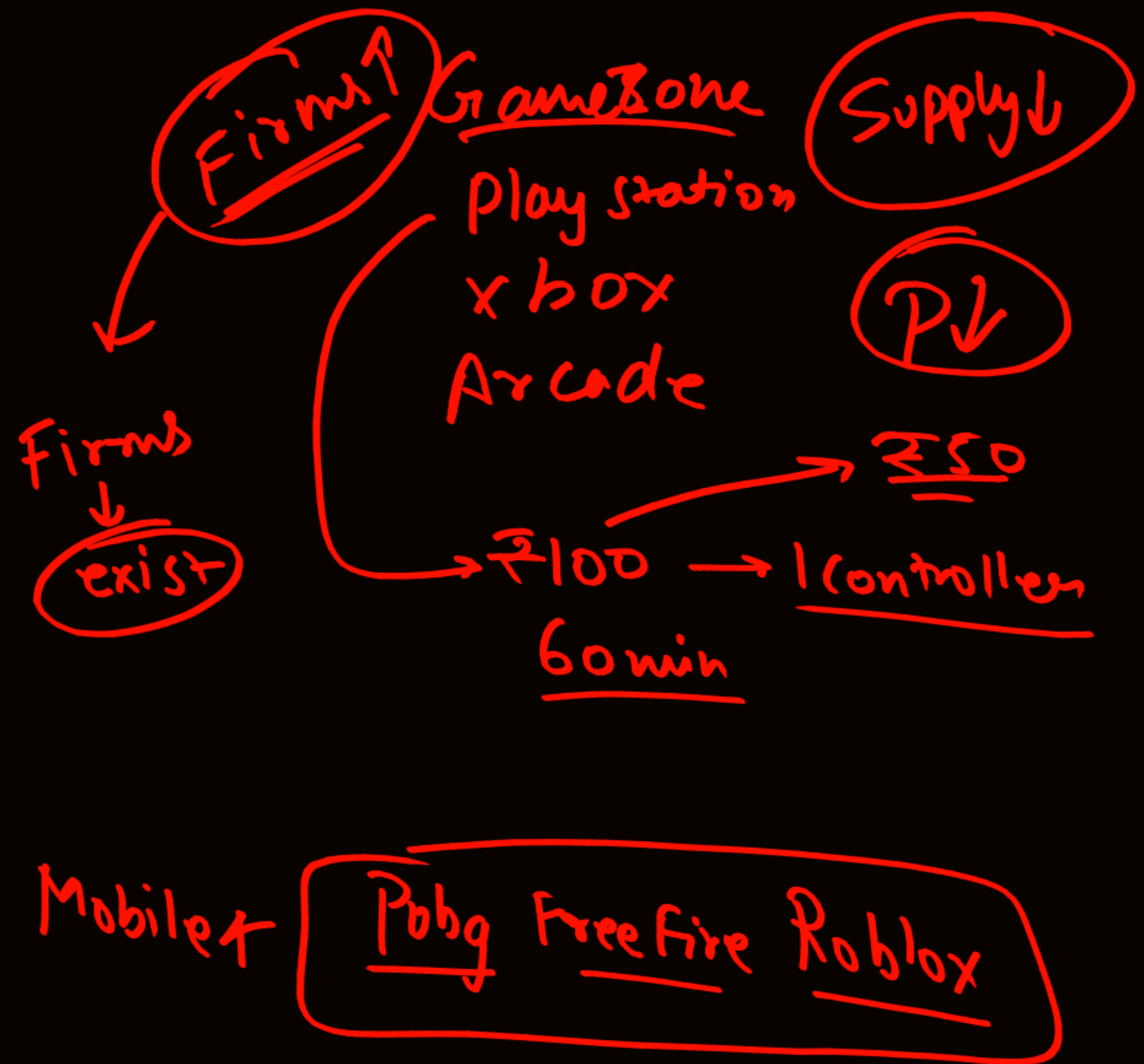
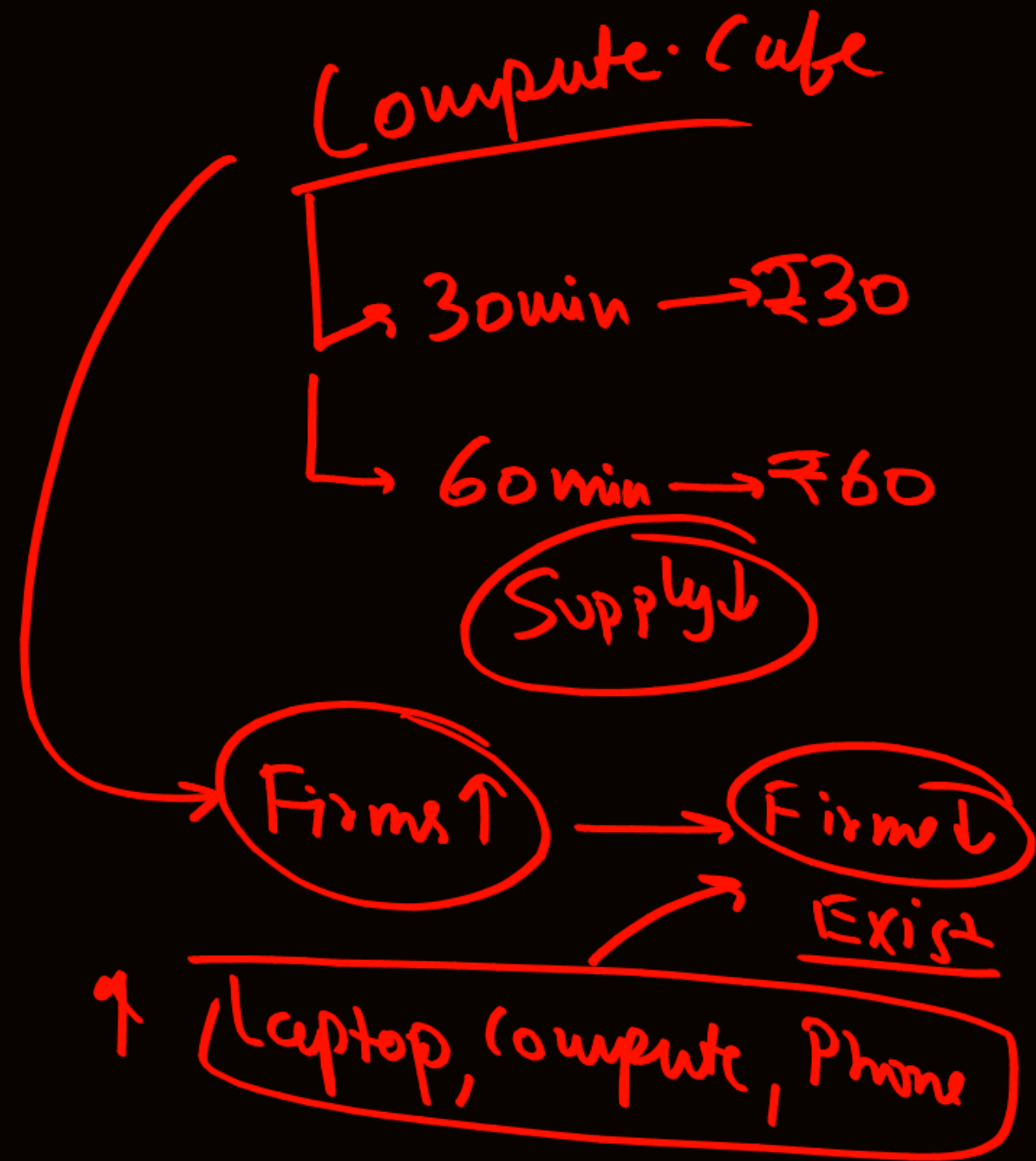
Change in stock

When the price of a good increases, the sellers are ready to supply more goods from their stocks. However, at a relatively lower price, the producers do not release big quantities from their stocks. They start increasing their inventories with a view that price may rise in near future.

Stock
Maintain

→ P↑ S↑
→ P↓ S↓

S↑ → Profit↑
S↓ → Profit↓



Exceptions to Law of Supply

Future Expectations

If sellers expect a fall in price in the future, then the law of supply may not hold true. In this situation, the sellers will be willing to sell more even at a lower price. However, if they expect the price to rise in the future, they would reduce the supply of the commodity, in order to supply the commodity later at a high price.



Agricultural Goods

The law of supply does not apply to agricultural goods as their production depends on climatic conditions. If, due to unforeseen changes in weather, the production of agricultural products is low, then their supply cannot be increased even at higher prices.



Perishable Goods

In case of perishable goods, like vegetables, fruits, etc., sellers will be ready to sell more even if the prices are falling. It happens because sellers cannot hold such goods for long.



Rare Articles

Rare, artistic and precious articles are also outside the scope of law of supply. For example, supply of rare articles like painting of Mona Lisa cannot be increased, even if their prices are increased.



Backward Countries

In economically backward countries, production and supply cannot be increased with rise in price due to shortage of resources.



AFRICA

low supply

14

Feb

PL $\uparrow$



Rose
₹5

vignt

Sani

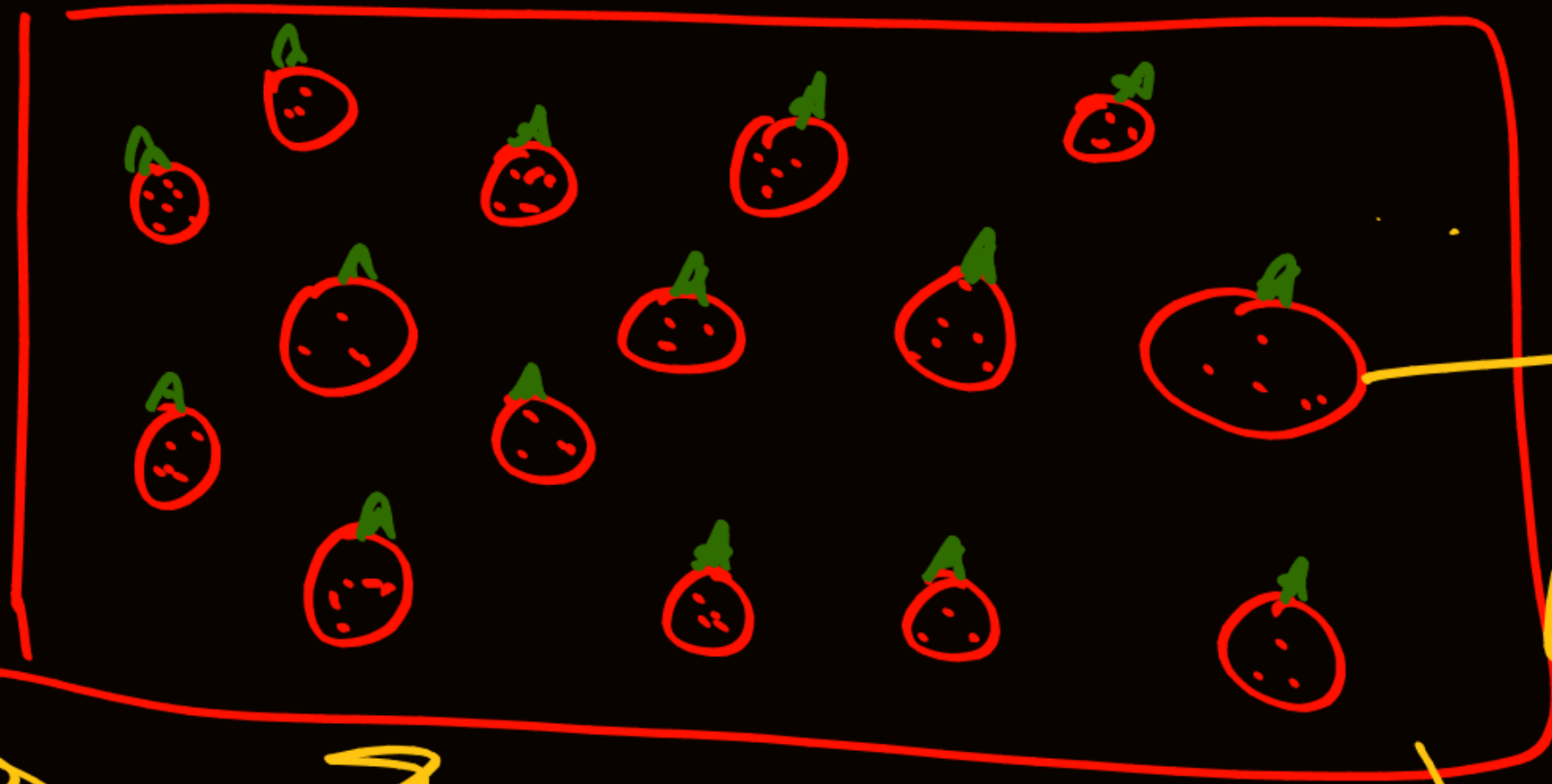
Sonari



Rose
₹100

Ambani

Soni Farm



Strawberries

Today
Seeds

winters

STX

₹1 lakh

₹10 lakh

₹1 cr.

DOUBT



**THANK
YOU**

Stay Connected

KEEP LEARNING